Higher Implicit Derivatives

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1 Setup

Let $F(x, y)$ be a smooth (say C^∞) function of two variables, and consider the level set

$$F(x, y) = C$$

for some constant C. Near any point where $F_y \neq 0$, the Implicit Function Theorem guarantees that this equation defines y as a function of x, say $y = y(x)$, at least locally. We are interested in computing the derivatives y', y'', y''', ... directly from F and its partial derivatives, without ever solving for $y(x)$ explicitly.

The method is always the same: differentiate the identity $F(x, y(x)) = C$ with respect to x, using the chain rule, and then solve algebraically for whichever derivative of y is new. Each differentiation introduces one new unknown (y', then y'', then y''', ...) and one new equation, so the system is always solvable as long as $F_y \neq 0$.

Throughout, subscripts on F denote partial derivatives: $F_x = \partial F/\partial x$, $F_{xy} = \partial^2 F/\partial x\, \partial y$, and so on. We assume enough smoothness that mixed partials commute (Clairaut's theorem), e.g. $F_{xy} = F_{yx}$.

2 First derivative

Differentiating $F(x, y(x)) = C$ once with respect to x:

$$F_x + F_y\, y' = 0 \quad \Longrightarrow \quad y' = -\frac{F_x}{F_y}.$$

This is the standard implicit differentiation formula. Nothing new yet, but it sets the pattern: every subsequent step is "differentiate the previous equation again, then solve for the new unknown."

3 Second derivative

Differentiate $F_x + F_y y' = 0$ once more with respect to x. The key point is that F_x and F_y are themselves functions of x and y, so differentiating them requires the chain rule again:

$$\frac{d}{dx}F_x = F_{xx} + F_{xy}\, y', \qquad \frac{d}{dx}F_y = F_{yx} + F_{yy}\, y'.$$

Combined with the product rule on the term $F_y y'$, this gives

$$F_{xx} + F_{xy}y' + \big(F_{yx} + F_{yy}y'\big)y' + F_y y'' = 0.$$

Using $F_{xy} = F_{yx}$ and collecting terms:

$$F_{xx} + 2F_{xy}y' + F_{yy}(y')^2 + F_y y'' = 0.$$

Solving for y'':

$$y'' = -\frac{F_{xx} + 2F_{xy}y' + F_{yy}(y')^2}{F_y}.$$

This expression still contains y'; substituting $y' = -F_x/F_y$ and clearing denominators gives the closed form

$$\boxed{y'' = -\frac{F_{xx}F_y^2 - 2F_{xy}F_xF_y + F_{yy}F_x^2}{F_y^3}}.$$

Connection to the bordered Hessian. The numerator is (up to sign) the determinant

$$\begin{vmatrix} F_{xx} & F_{xy} & F_x \\ F_{xy} & F_{yy} & F_y \\ F_x & F_y & 0 \end{vmatrix} = F_y^2F_{xx} - 2F_xF_yF_{xy} + F_x^2F_{yy},$$

which is exactly the numerator above. This matrix (the Hessian of F bordered by its gradient) is the same object that appears in second-derivative tests for constrained optimization and in formulas for the curvature of a level curve. The recurring appearance of this determinant is not a coincidence: y'', curvature, and the second-order behavior of Lagrange-multiplier problems are all measuring "how much F curves relative to how steeply it is changing," which is precisely what a bordered Hessian encodes.

4 Third derivative

The same procedure continues. Differentiate

$$F_{xx} + 2F_{xy}y' + F_{yy}(y')^2 + F_yy'' = 0$$

once more with respect to x, remembering that every partial of F again picks up a $(\,\cdot\,)_y \cdot y'$ term via the chain rule, and that $2F_{xy}y'$, $F_{yy}(y')^2$, and F_yy'' each need the product rule. Carrying this out and simplifying (using $F_{xy} = F_{yx}$ and $F_{xyy} = F_{yxy} = F_{yyx}$, etc.) yields

$$F_{xxx} + 3F_{xxy}y' + 3F_{xyy}(y')^2 + F_{yyy}(y')^3 + 3F_{xy}y'' + 3F_{yy}y'y'' + F_yy''' = 0,$$

so that

$$y''' = -\frac{F_{xxx} + 3F_{xxy}y' + 3F_{xyy}(y')^2 + F_{yyy}(y')^3 + 3F_{xy}y'' + 3F_{yy}y'y''}{F_y}.$$

As with y'', one can substitute the already-known expressions for y' and y'' to get a formula purely in terms of F's partial derivatives, but the resulting expression is long and rarely worth memorizing. The binomial-looking coefficients $(1, 3, 3, 1)$ are not an accident: they come from repeatedly applying the chain and product rules to F composed with $(x, y(x))$, which is structurally close to a Faà di Bruno / Leibniz expansion.

5 The general pattern

Every order follows the same three-step recipe:

(1) Start from the equation obtained at the previous order (at order 1, this is $F(x, y) = C$ itself).

(2) Differentiate the whole equation with respect to x, applying the chain rule to every term involving F or its partials (since they depend on y, which depends on x), and the product rule to every term already containing a derivative of y.

(3) Collect terms and solve algebraically for the new, highest-order derivative of y that appears. It will always show up linearly, multiplied by F_y, because each differentiation of $F(x, y(x))$ with respect to x produces exactly one term of the form $F_y \cdot$ (new derivative) from the chain rule applied to the y-dependence.

This is why the requirement $F_y \neq 0$ is exactly what is needed to solve for y', y'', y''', and so on: at every order, F_y is the coefficient of the new unknown.

6 Worked examples

Example 1: the unit circle

Let $F(x, y) = x^2 + y^2$ and $C = 1$, i.e. the unit circle $x^2 + y^2 = 1$.

Partials: $F_x = 2x$, $F_y = 2y$, $F_{xx} = 2$, $F_{yy} = 2$, $F_{xy} = 0$.

First derivative:

$$y' = -\frac{2x}{2y} = -\frac{x}{y}.$$

Second derivative, using the boxed formula:

$$y'' = -\frac{F_{xx}F_y^2 - 2F_{xy}F_xF_y + F_{yy}F_x^2}{F_y^3} = -\frac{2(2y)^2 - 0 + 2(2x)^2}{(2y)^3} = -\frac{8y^2 + 8x^2}{8y^3} = -\frac{x^2 + y^2}{y^3}.$$

Since $x^2 + y^2 = 1$ on the circle, this simplifies to

$$y'' = -\frac{1}{y^3}.$$

This matches the direct computation from $y = \sqrt{1 - x^2}$, and it has a clean geometric meaning: for the unit circle, curvature is constant, and y'' reflects that constancy once restricted to the curve.

Example 2: a curve with a mixed term

Let $F(x, y) = x^2 + xy + y^2$ and $C = 3$ (an ellipse, since the quadratic form is positive definite).

Partials: $F_x = 2x + y$, $F_y = x + 2y$, $F_{xx} = 2$, $F_{yy} = 2$, $F_{xy} = 1$.

First derivative:

$$y' = -\frac{2x + y}{x + 2y}.$$

Second derivative:

$$y'' = -\frac{2(x + 2y)^2 - 2(1)(2x + y)(x + 2y) + 2(2x + y)^2}{(x + 2y)^3}.$$

Expanding the numerator:

$$2(x + 2y)^2 = 2x^2 + 8xy + 8y^2, \qquad 2(2x + y)(x + 2y) = 4x^2 + 10xy + 4y^2,$$

$$2(2x+y)^2 = 8x^2 + 8xy + 2y^2,$$

so the numerator is

$$(2x^2+8xy+8y^2) - (4x^2+10xy+4y^2) + (8x^2+8xy+2y^2) = 6x^2+6xy+6y^2 = 6(x^2+xy+y^2).$$

Using $x^2 + xy + y^2 = 3$ on the curve, the numerator is 18, so

$$y'' = -\frac{18}{(x+2y)^3}.$$

This example shows the mixed-partial term F_{xy} doing real work: without the $-2F_{xy}F_xF_y$ term, the cancellation down to a multiple of F itself would not happen.

Example 3: a transcendental curve

Let $F(x, y) = \sin(x) + e^y - xy$ and $C = 1$, i.e. $\sin x + e^y - xy = 1$.

Partials: $F_x = \cos x - y$, $F_y = e^y - x$, $F_{xx} = -\sin x$, $F_{yy} = e^y$, $F_{xy} = -1$.

First derivative:

$$y' = -\frac{\cos x - y}{e^y - x}.$$

Second derivative, directly from the boxed formula (no further simplification is available here, since the curve is not defined by a homogeneous polynomial):

$$y'' = -\frac{(-\sin x)(e^y - x)^2 - 2(-1)(\cos x - y)(e^y - x) + (e^y)(\cos x - y)^2}{(e^y - x)^3}.$$

This can be evaluated at any specific point (x_0, y_0) on the curve by plugging in numbers; it illustrates that the method works identically whether or not F is polynomial, since nothing in the derivation used polynomiality — only differentiability.

7 Self-check questions

1. Why does the coefficient of the newest derivative of y (i.e. y' in the first equation, y'' in the second, and so on) always turn out to be exactly F_y, at every order?
2. Derive y'' from scratch for $F(x, y) = x^3 - 3xy + y^3$, $C = 0$ (the folium of Descartes), without looking at the boxed formula. Then check your answer against the boxed formula.
3. The boxed formula for y'' can be written as a 3×3 determinant divided by F_y^3. What goes wrong (in terms of that determinant, or in terms of the original derivation) if $F_y = 0$ at a point on the curve? What does this correspond to geometrically?
4. In the derivation of y''', coefficients $1, 3, 3, 1$ appeared. Where exactly do they come from? (Hint: track how many times the product rule is invoked on terms that already contain a y' or y'' factor, versus how many times the chain rule is invoked on partials of F.)
5. Suppose $F(x, y) = f(x) + g(y)$ for some single-variable functions f, g (i.e. F is "separable"). Show that the general boxed formula for y'' collapses to something much simpler. Can you explain why separability should make the computation easier, structurally?
6. If $F(x, y) = C$ implicitly defines x as a function of y instead (i.e. you solve for $x(y)$), what is the analogous formula for $x''(y)$? What symmetry do you expect between the $y''(x)$ and $x''(y)$ formulas, and does the boxed formula actually have that symmetry?

7. For Example 1 (the unit circle), the formula for y'' simplified using the constraint $x^2+y^2=1$. Explain in general terms why it is valid to substitute the original constraint $F(x,y)=C$ back into a formula for y'', even though y'' was derived by differentiating, not by using the constraint directly.
8. Explain, in your own words and without more computation, why y''' needs $F_y \neq 0$ at exactly the same points where y'' and y' do — i.e. why no new nondegeneracy condition appears at higher orders.